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Machine Learning 2024-2045

Homework 3 - Group 127

I. - Pen-and-Paper

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Pen-and-Paper

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D	y_1	y_2	y_{num}	y_{class}
x_1	1	1	1,25	B
x_2	1	3	7	A
x_3	3	2	2.7	C
x_4	3	3	3,2	A
x_5	2	4	5.5	B

$\phi(y_1, y_2) = y_1 \times y_2$

Transforming input space

$\varphi(x_1) = \varphi(1,1) = 1 \times 1 = 1$

$\varphi(x_2) = \varphi(1,3) = 1 \times 3 = 3$

$\varphi(x_3) = \varphi(3,2) = 3 \times 2 = 6$

$\varphi(x_4) = \varphi(3,3) = 3 \times 3 = 9$

$\varphi(x_5) = \varphi(2,4) = 2 \times 4 = 8$

Transformed input matrix:

$$\Phi = \begin{bmatrix} 1 \\ 3 \\ 6 \\ 9 \\ 8 \end{bmatrix}$$

Target values: $Y = \begin{bmatrix} 1,25 \\ 7 \\ 2,7 \\ 3,2 \\ 5,5 \end{bmatrix}$

OLS Design matrix: $X = \begin{bmatrix} 1 & 1 \\ 1 & 3 \\ 1 & 6 \\ 1 & 9 \\ 1 & 8 \end{bmatrix}$

OLS closed form solution: $w = (X^T X)^{-1} X^T Y$

$X^T \rightarrow$ transpose of X

$(X^T X)^{-1} \rightarrow$ inverse of $X^T X$

To calculate the resulting vector, we used a small python code and the linalg.pinv function (from the hint).

Here is the code:

```
import numpy as np
X = np.array([1,1],[1,3],[1,6],[1,9],[1,8]])
Y = np.array([1,25, 7.00, 2.70, 3.20, 5.50])
w = np.linalg.pinv(X.T @ X) @ X.T @ Y
```

$$w = \begin{bmatrix} 3.31593 \\ 0.11372 \end{bmatrix}$$

OLS regression model:

$\hat{y}_{num} = 3.31593 + 0.11372 \times \varphi(y_1, y_2)$

Final regression model:

$\hat{y}_{num} = 3.31593 + 0.11372 \cdot (y_1 \times y_2)$

```python
import numpy as np

# Design matrix X
X = np.array([[1, 1],
              [1, 3],
              [1, 6],
              [1, 9],
              [1, 8]])

# Target vector Y (y_num)
Y = np.array([1.25, 7.00, 2.70, 3.20, 5.50])

# Compute the OLS solution
w = np.linalg.pinv(X.T @ X) @ X.T @ Y
```

Pen-and-paper

(2) Ridge Regression

$$J(\hat{w}) = \sum (y - \hat{y})^2 + \lambda \sum w_i^2$$

$\lambda \rightarrow$ penalty factor

$w_i \rightarrow$ weights

The larger the λ, the stronger the penalty

From the previous exercise: $\Phi = \begin{bmatrix} 1 \\ 3 \\ 6 \\ 9 \\ 8 \end{bmatrix}$ $X = \begin{bmatrix} 1 & 1 \\ 1 & 3 \\ 1 & 6 \\ 1 & 9 \\ 1 & 8 \end{bmatrix}$ $Y = \begin{bmatrix} 1,25 \\ 7.00 \\ 2,70 \\ 3.20 \\ 5,50 \end{bmatrix}$

- Ridge formula:

$$\hat{w}_{ridge} = (X^T X + \lambda I)^{-1} \cdot X^T \cdot Y$$

- Using the code, and $\lambda = 1$,

we got the intercept bias = 1,81809

slope = 0,32376

$$\hat{y}_{num}^{ridge} = 1,81809 + 0,32376 \times \varphi(y_1, y_2)$$

we used the same code as the previous exercise, but ~~[illegible]~~ added these lines:

- `lambda = 1` # beggining of code
- `I = n.p.eye(X.shape[1])` # after the arrays
- `w_ridge = np.linalg.pinv(X.T @ X + lambda*I) @X.T @ y`

- When penalizing the intercept in Ridge regression, the intercept drops from 3.31593 to 1.81809, while the slope increases from 0.11372 to 0.32376, as the model compensates for the smaller intercept by assigning more weight to the feature coefficient.

- Regularization in Ridge regression significantly reduces the intercept and increases the slope, redistributing the model's complexity from the intercept to the feature coefficients.

```python
import numpy as np

# Define the lambda (penalty) value
lambda_ = 1

# Design matrix X (with intercept) and target vector Y
X = np.array([[1, 1],
              [1, 3],
              [1, 6],
              [1, 9],
              [1, 8]])

Y = np.array([1.25, 7.00, 2.70, 3.20, 5.50])

# Identity matrix for regularization
I = np.eye(X.shape[1])

# Ridge regression closed-form solution
w_ridge = np.linalg.pinv(X.T @ X + lambda_ * I) @ X.T @ Y
w_ridge
```

Pen-and-Paper

(3) Root Mean Square Error (RMSE): $\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_{true} - y_{pred})^2}$

- $n6 = (2,2)$, $y6 = 07$, $\phi(n6) = 2 \times 2 = 4$
- $n7 = (1,2)$, $y7 = 1,1$, $\phi(n7) = 1 \times 2 = 2$
- $n8 = (5,1)$, $y8 = 2,2$, $\phi(n8) = 5 \times 1 = 5$

Predictions For Training Data (OLS): $X = \begin{bmatrix} 1 \\ 3 \\ 6 \\ 9 \\ 8 \end{bmatrix}$, from exercise 1 $y = [1,25; 7; 2,7; 3,2; 5,5]$

$\hat{y}^{ols}_{num} = 3.37593 + 0.11372 \times X_i$

$X_1 = 1$; $\hat{y} = 3,37593 + 0,11372 \times 1 = 3,42965$

$X_2 = 3$; $\hat{y} = 3,37593 + 0,11372 \times 3 = 3,65709$

$X_3 = 6$; $\hat{y} = $ " " " $\times 6 = 4,01625$

$X_4 = 9$; $\hat{y} = $ " " " $\times 9 = 4,37541$

$X_5 = 8$; $\hat{y} = $ " " " $\times 8 = 4,26169$

Predictions for training Data (Ridge):

$\hat{y}^{ridge}_{num} = 1,81809 + 0,32376 \times X_i$, from exercise 2

$X_1 = 1$; $\hat{y} = 1,81809 + 0.32376 \times 1 = 2,14185$

$X_2 = 3$; $\hat{y} = $ " " " $\times 3 = 2,78937$

$X_3 = 6$; $\hat{y} = $ " " " $\times 6 = 3,76065$

$X_4 = 9$; $\hat{y} = $ " " " $\times 9 = 4,73193$

$X_5 = 8$; $\hat{y} = $ " " " $\times 8 = 4,40817$

OLS Training RMSE

$Errors = [y_{true} - y_{pred}] = [1.25 - 3,42965; 7 - 3,65709; 2.7 - 4,01625; 3.2 - 4,37541; 5,5 - 4,26169] =$

$= [-2,17965; 3.34291; -1,31625; -1,17541; 1,23831]$

$Errors^2 = [4,75089; 11,17608; 1,73252; 1.38159; 1,53342]$

$\sum Errors^2 = 20,57449$

$RMSE^{ols}_{training} = \sqrt{\frac{20,57449}{5}} = 2.02750$

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(3) cont.

Ridge Training RMSE:

Errors = [1,15 − 2,14185 ; 7 − 2,78937 ; 2,7 − 3,76065 ; 3,2 − 4,73793 ; 5,5 − 4,40817]

= [−0,89185 ; 4,21063 ; −1,06065 ; −1,53193 , 1,09183]

Errors² = [0,7954 ; 17,73339 ; 1,1254 ; 2,34787 ; 1,1911]

ΣErrors² = 23,1927

$RMSE^{ridge}_{training} = \sqrt{\frac{23,1927}{5}} = 2,15354$

Predictions for test Data (OLS):

n6; $\hat{y}$ = 3,31593 + 0,11372 × 4 = 3.77081

n7; $\hat{y}$ = " " " × 2 = 3.54337

n8; $\hat{y}$ = " " " × 5 = 3.88853

Predictions for test Data (Ridge)

n6; $\hat{y}$ = 1,87809 + 0,32376 × 4 = 3,17313

n7; $\hat{y}$ = " " " × 2 = 2,46561

n8; $\hat{y}$ = " " " × 5 = 3,43789

- OLS Test RMSE

Errors = [0,7 − 3,77081 ; 1,1 − 3,54337 ; 2,2 − 3.88453] = [−3,07081 ; −2,44337 ; −1,68453]

Errors² = [9,42988 ; 5,96805 ; 2,83864]

ΣErrors² = 18,23657

$RMSE^{OLS}_{test} = \sqrt{\frac{18,23657}{3}} = 2,46560$

Ridge Test RMSE

Errors = [0,7 − 3,17313 ; 1,1 − 2,46561 ; 2,2 − 3,43789] = [−2,47313 ; −1,36561 ; −1,23789]

Errors² = [5,8252 ; 1,86489 ; 1,53238]

ΣErrors² = 9,22247

$RMSE^{ridge}_{test} = \sqrt{\frac{9,22247}{3}} = 1,75290$

Conclusion

- Ridge regression has a significantly lower RMSE on the test set compared to OLS. This aligns with the expected behaviour, Ridge regularization improves generalization by penalizing the coefficients, which helps the model perform better on unseen data.

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③ Cont.

OLS performs slightly better on the training set than Ridge, which is expected because OLS minimizes the training error without any regularization. Ridge, however, sacrifices some training performance to prevent overfitting, which is reflected in the slighty higher error in training.

④ Steps:

1. Forward Pass
2. Compute loss
3. Backward Pass
4. Update Weights and Biases

- Input: $x_1 = [1, 1]$
- First layer Computation: $z^{[1]} = W^{[1]} \cdot x_1 + b^{[1]}$

$$z^{[1]} = \begin{bmatrix} 0.1 & 0.1 \\ 0.1 & 0.2 \\ 0.2 & 0.1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0 \\ 0.1 \end{bmatrix} \Rightarrow$$

$$\Rightarrow z^{[1]} = \begin{bmatrix} 0.1 \cdot 1 + 0.1 \cdot 1 \\ 0.1 \cdot 1 + 0.2 \cdot 1 \\ 0.2 \cdot 1 + 0.1 \cdot 1 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0 \\ 0.1 \end{bmatrix} \Rightarrow z^{[1]} = \begin{bmatrix} 0.2 \\ 0.3 \\ 0.3 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0 \\ 0.1 \end{bmatrix} \Rightarrow$$

$$\Rightarrow z^{[1]} = \begin{bmatrix} 0.3 \\ 0.3 \\ 0.4 \end{bmatrix}$$

- Second Layer Computation: $z^{[2]} = W^{[2]} \cdot z^{[1]} + b^{[2]}$

$$z^{[2]} = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0.3 \\ 0.3 \\ 0.4 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.3 \cdot 1 + 2 \cdot 0.3 + 1 \cdot 0.3 \\ 1 \cdot 0.3 + 1 \cdot 0.3 + 1 \cdot 0.3 \\ 1 \cdot 0.4 + 1 \cdot 0.4 + 1 \cdot 0.4 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1,7 \\ 1,3 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2,7 \\ 2,3 \\ 2 \end{bmatrix}$$

- Applying Softmax: $\text{Softmax}(z_L^{[out]}) = \dfrac{e^{z_L^{[out]}}}{\sum_{l=1}^{|L|} e^{z_l^{[out]}}}$

$$e^{z^{[2]}} = \begin{bmatrix} e^{2,7} \\ e^{2,3} \\ e^{2} \end{bmatrix} = \begin{bmatrix} 14,8797 \\ 9,9742 \\ 7,3891 \end{bmatrix}$$

$$\sum_{l=1}^{3} e^{z_l^{[out]}} = e^{2,7} + e^{2,3} + e^{2} = 32,2429$$

Predicted probabilies for classes A, B, C:

$$\hat{y}_A = \frac{14,8797}{32,2429} = 0,4614 \qquad \hat{y}_B = \frac{9,9742}{32,2429} = 0,3093 \qquad \hat{y}_C = \frac{7,3891}{32,2429} = 0,2291$$

Cross-Entropy Loss

for x_1, the true label is class A, so target vector t is: $t = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$L = -\sum_{L=1}^{|L|} t_L \cdot \log(\hat{y}_L)$, this simplifies to: $L = -\log(\hat{y}_A) = -\log(0,4614) = 0,7731$

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(4) cont.

Gradient at output layer

$$\frac{\partial L}{\partial z_i^{[2]}} = \hat{y}_i - y_i$$

For each class:

$$\frac{\partial L}{\partial z_A^{[2]}} = 0,4615 - 1 = -0,5385$$

$$\frac{\partial L}{\partial z_B^{[2]}} = 0,3093 - 0 = 0,3093$$

$$\frac{\partial L}{\partial z_C^{[2]}} = 0,2291 - 0 = 0,2291$$

Gradient for second layer weights and biases

weights: $\frac{\partial L}{\partial w_{ij}^{[2]}} = \frac{\partial L}{\partial z_i^{[2]}} \times z_j^{[1]}$ biases: $\frac{\partial L}{\partial b_i^{[2]}} = \frac{\partial L}{\partial z_i^{[2]}}$

$$\frac{\partial L}{\partial z^{[2]}} = \begin{bmatrix} -0,5385 \\ 0,3093 \\ 0,2291 \end{bmatrix}$$

$$z^{[1]} = \begin{bmatrix} 0,3 \\ 0,3 \\ 0,1 \end{bmatrix}$$

- For class A (output $z_A^{[2]}$):

$$\frac{\partial L}{\partial w_{A1}^{[2]}} = -0,5385 \times 0,3 = -0,16155$$

$$\frac{\partial L}{\partial w_{A2}^{[2]}} = -0,5385 \times 0,3 = -0,16155$$

$$\frac{\partial L}{\partial w_{A3}^{[2]}} = -0,5385 \times 0,1 = -0,2754$$

- For class B (output $z_B^{[2]}$):

$$\frac{\partial L}{\partial w_{B1}^{[2]}} = 0,3093 \times 0,3 = 0,09279$$

$$\frac{\partial L}{\partial w_{B2}^{[2]}} = 0,3093 \times 0,3 = 0,09279$$

$$\frac{\partial L}{\partial w_{B3}^{[2]}} = 0,3093 \times 0,1 = 0,12372$$

For class C (output $z_C^{[2]}$):

$$\frac{\partial L}{\partial w_{C1}^{[2]}} = 0,2291 \times 0,3 = 0,06873$$

$$\frac{\partial L}{\partial w_{C2}^{[2]}} = 0,2291 \times 0,3 = 0,06873$$

$$\frac{\partial L}{\partial w_{C3}^{[2]}} = 0,09164 = 0,2291 \times 0,4$$

Gradient Matrix for $W^{[2]}$:

$$\frac{\partial L}{\partial W^{[2]}} = \begin{bmatrix} -0,16155 & -0,16155 & -0,2754 \\ 0,09279 & 0,09279 & 0,12372 \\ 0,06873 & 0,06873 & 0,09164 \end{bmatrix}$$

Gradient Matrix for $b^{[2]}$: $\frac{\partial L}{\partial b^{[2]}} = \begin{bmatrix} -0,5385 \\ 0,3093 \\ 0,2291 \end{bmatrix}$

4 cont.

Backpropagation to the first layer

$$\frac{\partial L}{\partial z^{[2]}} = \begin{bmatrix} -0,5385 \\ 0,3093 \\ 0,2291 \end{bmatrix}$$

$$\frac{\partial L}{\partial z_j^{[1]}} = \sum_i \frac{\partial L}{\partial z_i^{[2]}} \cdot W_{ij}^{[2]}$$

$z_1^{[1]}$: $\frac{\partial L}{\partial z_1^{[1]}} = (-0,5385 \times 1) + (0,3093 \times 1) + (0,2291 \times 1) = -0,0001$

$z_2^{[1]}$: $\frac{\partial L}{\partial z_2^{[1]}} = (-0,5385 \times 2) + (0,3093 \times 2) + (0,2291 \times 1) = -0,2293$

$z_3^{[1]}$: $\frac{\partial L}{\partial z_3^{[1]}} = (-0,5385 \times 2) + (0,3093 \times 1) + (0,2291 \times 1) = -0,5386$

$$\frac{\partial L}{\partial z^{[1]}} = \begin{bmatrix} -0,0001 \\ -0,2293 \\ -0,5386 \end{bmatrix}$$

Gradients for the first layer weights $W^{[1]}$: $\frac{\partial L}{\partial W_{ij}^{[1]}} = \frac{\partial L}{\partial z_i^{[1]}} \times x_j$

$x_j = (1,1) = x_1$ $\quad \frac{\partial L}{\partial W^{[1]}} = \begin{bmatrix} -0,0001 \\ -0,2293 \\ -0,5386 \end{bmatrix} \times [1,1] = \begin{bmatrix} -0,0001 & -0,0001 \\ -0,2293 & -0,2293 \\ -0,5386 & -0,5386 \end{bmatrix}$

Gradients for the first layer biases $b^{[1]}$: $\frac{\partial L}{\partial b^{[1]}} = \begin{bmatrix} -0,0001 \\ -0,2293 \\ -0,5386 \end{bmatrix}$

Update weights and biases:

SGD using $\eta = 0.1$

$W_{new} = W_{old} - \eta \cdot \frac{\partial L}{\partial W}$

$b_{new} = b_{old} - \eta \cdot \frac{\partial L}{\partial b}$

$b^{[2]}$: $b_1^{[2]} = 1 - 0,1 \times (-0,5385) = 1,05385$

$b_2^{[2]} = 1 - 0,1 \times (0,3039) = 0,96907$

$b_3^{[2]} = 1 - 0,1 \times 0,2291 = 0,97709$

$W^{[2]}$: $W_{11}^{[2]} = 1 - 0,1 \times (-0,16155) = 1,016155$

$W_{12}^{[2]} = 2 - 0,1 \times (-0,16155) = 2,016155$

$W_{13}^{[2]} = 2 - 0,1 \times (-0,21554) = 2,02154$

$W_{21}^{[2]} = 0,990721$

$W_{22}^{[2]} = 1,990721$

$W_{23}^{[2]} = 0,987628$

$W_{31}^{[2]} = 0,993127$

$W_{32}^{[2]} = 0,993127$

$W_{33}^{[2]} = 0,990836$

$$b^{[2]}_{new} = \begin{bmatrix} 1,05385 \\ 0,96907 \\ 0,97709 \end{bmatrix}$$

$$W^{[2]}_{new} = \begin{bmatrix} 1,016155 & 2,016155 & 2,02154 \\ 0,990721 & 1,990721 & 0,987628 \\ 0,993127 & 0,993127 & 0,990836 \end{bmatrix}$$

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(4) cont.

for the $W^{[1]}_{new}$ and $b^{[1]}_{new}$ we followed the same steps for $W^{[2]}_{new}$ and $b^{[2]}_{new}$

$W^{[1]}$:

$W_{11}^{[1]} = 0,1 - 0,1 \times (-0,0001) = 0,10001$

$W_{12}^{[1]} = 0,1 - 0,1 \times (-0,0001) = 0,10001$

$W_{21}^{[1]} = 0,122493$

$W_{22}^{[1]} = 0,122493$

$W_{31}^{[1]} = 0,125386$

$W_{32}^{[1]} = 0,125386$

$$W^{[1]}_{new} = \begin{bmatrix} 0,10001 & 0,10001 \\ 0,122493 & 0,122493 \\ 0,125386 & 0,125386 \end{bmatrix} \qquad b^{[1]}_{new} = \begin{bmatrix} 0,10001 \\ 0,122493 \\ 0,125386 \end{bmatrix}$$